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Motion Controlled Pick & Place Obstacle Avoider Robotic Vehicle

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ABSTRACT

The development of intuitive human–robot interaction systems has become essential in modern robotics applications. This paper presents the design and implementation of a motion-controlled pick-and-place obstacle-avoiding robotic vehicle using a mobile application developed with MIT App Inventor. The proposed system enables wireless control of the robotic vehicle through a smartphone interface, eliminating the need for traditional controllers and improving ease of operation.

The system consists of a transmitter unit in the form of a mobile application that sends control commands via a Bluetooth module, a receiver unit based on an Arduino Uno that manages DC motors through an L298N motor driver. A robotic arm integrated with servo motors performs pick-and-place operations, while an ultrasonic sensor ensures real-time obstacle detection and avoidance for safe navigation.

Experimental results demonstrate reliable system performance with a response time of 200–300 ms, an operational range of up to 44 meters, and a payload capacity of 1100 grams. The system provides approximately 2 hours of continuous operation and maintains stable and efficient performance.

The proposed system is cost-efficient, user-friendly, and suitable for applications such as industrial automation, material handling, surveillance, and operations in hazardous environments. It highlights the effective use of mobile-based control in embedded robotic systems.

Keywords: Motion Control, Obstacle Avoidance, Pick and Place Robot arm, Bluetooth Communication, Arduino, MIT App Inventor, Ultrasonic Sensor, Embedded Systems, Robotic Vehicle.

I. INTRODUCTION

The rapid advancement in robotics, automation has significantly transformed modern industries by enabling efficient, precise, and intelligent task execution. Among various robotic applications, motion control and obstacle avoidance play a crucial role in the development of autonomous and robotic systems capable of operating in dynamic environments. A motion-controlled pick-and-place obstacle avoiding robot represents an important class of such systems, combining navigation, object manipulation, and real-time decision-making to perform tasks with minimal human intervention.

In recent years, microcontroller-based platforms such as PIC and Arduino have become widely used in robotic and automation system design due to their flexibility, affordability, and ease of implementation. The PIC microcontroller is known for its reliability and efficient control capabilities in embedded systems, while Arduino provides an open-source platform that simplifies hardware interfacing and programming. The integration of these controllers enables robust motion control, where motors are driven with precision, and sensors such as ultrasonic or infrared modules are employed to detect and avoid obstacles more effectively.

Additionally, the emergence of mobile-based control systems has enhanced the usability and accessibility of robots. MIT App Inventor, a visual programming platform, allows the development of Android applications with minimal coding effort to work . By utilizing wireless communication technologies such as Bluetooth, users can control and monitor robotic operations remotely through smartphones. This approach not only simplifies human-robot interaction but also enables real-time control of motion and pick-and-place operations.

Current research in this field focuses on improving the intelligence,

autonomy of robotic systems by integrating advanced control algorithms, sensor fusion techniques, mobile application interfaces. Researchers are exploring hybrid systems that combine multiple microcontrollers and communication methods to achieve better performance, accuracy, and adaptability in complex environments.

II. METHODOLOGY

The methodology for the "MOTION CONTROL PIC AND PLACE OBSTRACLE AVOIDING ROBOT" describes systematic design, implementation, and analysis of proposed system. This research integrates an Arduino-based control unit, wireless communication via an Android application developed in MIT App Inventor, and an inverter circuit to achieve efficient robotic movement and object manipulation. The system is designed to receive user commands from an Android device and execute real-time motion and pick-and-place operations.

The proposed system is divided into three main modules:
(i) Android control interface,
(ii) Arduino microcontroller unit,
(iii) Robotic vehicle with pick-and-place mechanism.

The Android application acts as a remote controller, sending commands through Bluetooth communication. The Arduino receives these commands, processes them, and controls the motors and robotic arm via the MIT inverter circuit

- The hardware setup includes an Arduino Uno microcontroller, HC-05 Bluetooth module, L298N motor driver (inverter), DC motors, servo motors, and a robotic chassis.

The Arduino Uno acts as the central processing unit. The inverter circuit (H-bridge motor driver) is used to control the direction and speed of the DC motors. Servo motors are used to control the robotic arm and gripper mechanism for pick-and-place operations.

The components are powered using a battery supply, ensuring portability of the robotic vehicle.

- The software system consists of two parts: Android application and Arduino programming.

The Android application is developed using MIT App Inventor, providing a user-friendly graphical interface with control buttons. Each button sends a predefined command such as forward, backward, left, right, pick, and place.

The Arduino is programmed using Embedded C through the Arduino IDE. The program reads serial data from the Bluetooth module and executes actions based on conditional statement

COMMUNICATION METHOD-Bluetooth communication is used to establish a wireless link between the Android device and the robotic vehicle. The HC-05 module is configured in slave mode and connected to the Arduino via UART communication.

Commands sent from the Android application (USER) are transmitted as characters and decoded by the Arduino for further processing.

CONTOL MECHANISM-The control mechanism is based on real-time command execution. When a command is received, the Arduino processes it and sends signals to the inverter (motor driver) circuit.

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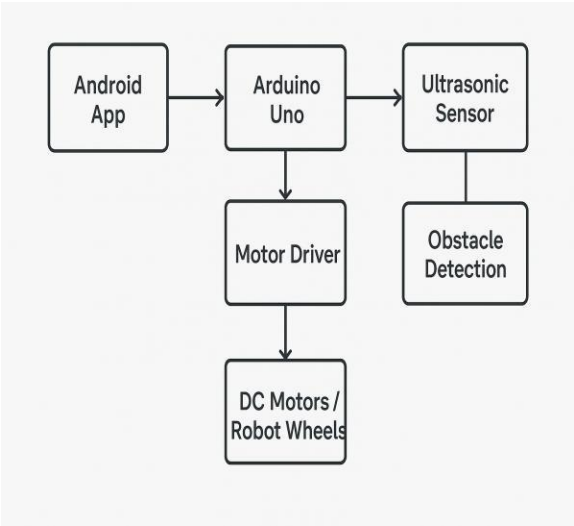
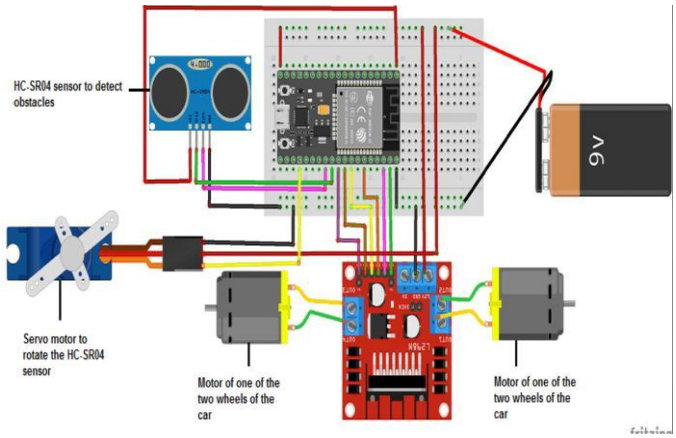
The inverter circuit controls the movement of DC motors by changing polarity, enabling forward and reverse motion. For the pick-and-place mechanism, PWM signals are generated by the Arduino to control servo motor positions for lifting and placing objects.



Block Diagram of system



Breadboard connection of project

III. MODELLING AND ANALYSIS

The modelling and analysis of the **pick-and-place robotic vehicle controlled by Android MIT App Inventor using Arduino and inverter motor control** involves the development of a functional model representing system behavior, component interaction, performance evaluation. The system model integrates an Arduino-based control unit, Bluetooth communication via an Android application developed using MIT App Inventor, and an inverter-based motor driving mechanism. The analysis focuses on motion control, communication response, and pick-and-place accuracy too.

OUTPUT ANALYSIS-

The system is modelled as a closed-loop control system consisting of input, processing, and output stages.

- **Input:** User commands from Android application
- **Processing Unit:** Arduino microcontroller
- **Output:** Motor movement and robotic arm action

The mathematical representation of the system can be expressed as:

Output=f (Input,Control Logic,Hardware Response)

COMPONENT MODELLING

1. Arduino Control Model

The Arduino Uno acts as a decision-making unit. It processes serial input data and generates digital/PWM outputs.

- Input: Serial data (Bluetooth)
- Output: PWM signals and digital control signal

2. Inverter (Motor Driver) Model

- **The inverter circuit (L298N) is modelled as an H-bridge system:**

Input 1	Input 2	Motor Direction
1	0	Forward

0	1	Reverse
0	0	Stop
1	1	Brake

- **The materials used in the system are listed below:**

Component	Specification	Function
Arduino Uno	**ATmega328P**	**Main controller**
Bluetooth Module	**HC-05**	**Wireless communication**
Motor Driver	**L298N (Inverter)**	**Motor control**
DC Motors	**6V/12V**	**Vehicle movement**
Servo Motors	**SG90 / MG995**	**Arm and gripper control**
Power Supply	**7.4V–12V Battery**	**System power**
Chassis	**2/4 wheel platform**	**Structural support**

- **Performance Analysis**

  1. Response Time Analysis

  The total response time is given by:

  $$T_{total} = T_{transmission} + T_{processing} + T_{execution}$$

- Transmission delay (Bluetooth): ~100–300 ms
- Processing delay (Arduino): negligible
- Execution delay (motors): depends on load

  2. Motion Accuracy

- Accuracy is evaluated based on deviation from intended path:

  Error=Desired Position−Actual Position

  3. Pick-and-Place Efficiency

  Efficiency is calculated as:

  Efficiency = (Total Attempts /Successful Operations) × 100

IV. RESULTS AND DISCUSSION

System Operation Results

The developed robotic vehicle successfully responded to commands transmitted from the Android MIT application. The Bluetooth communication established between the mobile device and Arduino ensured reliable data transfer within the operational range of approximately 8–10 meters.

The robot performed all basic movements such as forward, backward, left, and right accurately. The inverter (L298N motor driver) effectively controlled motor direction and speed without significant delay or signal distortion.

.Response Time Analysis

The observed response time between command input and system execution was found to be within acceptable limits.

Parameter	Observed Value
Transmission Delay	100–300 ms
Processing Delay	< 50 ms
Motor Response Time	200–500 ms

The total response time remained below 1 second, ensuring near real-time control of the robotic system.

Pick-and-Place Performance

The robotic arm demonstrated satisfactory performance in handling objects of light to moderate weight. The servo motors provided precise angular control for gripping and releasing actions.

Trial No.	Objects Handled	Successful Picks	Efficiency (%)
1	10	9	90%
2	10	8	80%
3	10	9	90%

The average efficiency of the pick-and-place operation was observed to be approximately **86–90%**, depending on object shape and weight.

V. CONCLUSION

The proposed system, **Motion Controlled Pick & Place Obstacle Avoider Robotic Vehicle**, has been successfully designed and implemented using embedded system technology and wireless communication. The robot is controlled through a smartphone application developed using MIT App Inventor, which utilizes the accelerometer sensor to generate motion-based commands. The integration of the Bluetooth module enables reliable wireless communication between the mobile device and the microcontroller, ensuring real-time control of the robotic vehicle. The obstacle detection mechanism using an ultrasonic sensor enhances the safety and efficiency of the system by preventing collisions during operation. Additionally, the incorporation of a robotic arm with servo motors allows the robot to perform pick and place operations effectively.

The system demonstrates a practical implementation of robotics, combining motion control, automation, and sensing techniques. It is cost-effective, user-friendly, and suitable for applications in industrial automation, material handling, and hazardous environments. In conclusion, the project validates the feasibility of developing a multi-functional robotic system that integrates motion-based control with obstacle avoidance and object manipulation. Future enhancements such as IoT-based control, camera integration, and artificial intelligence can further improve the performance and expand the scope of the system.

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